Boson14 Planted-Clique Benchmark

Variant & Amplitude Sweep on Motzkin–Straus StQP

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Motzkin–Straus in the scaled simplex

Given an undirected graph $G = (V, E)$ with adjacency A, the *standard quadratic program* over the scaled simplex is

$$\max_{y \geq 0,\, \sum_i y_i = R} \; g(y) \;=\; \tfrac{1}{2}\, y^\top A\, y.$$

The Motzkin–Straus theorem gives the optimum as a function of the clique number ω:

$$g^*(\omega) \;=\; \frac{R^2}{2}\left(1 - \frac{1}{\omega}\right), \qquad \omega \;=\; \mathrm{round}\left(\frac{R^2}{R^2 - 2\, g^*}\right).$$

Boson14 maximises $y^\top A\, y = 2\, g(y)$ under the sum constraint R, so the relation between the reported hardware "energy" and the scaled objective is

$$\texttt{energy_list}[i] \;=\; y_i^\top A\, y_i \;=\; 2\, g(y_i).$$

Experimental design

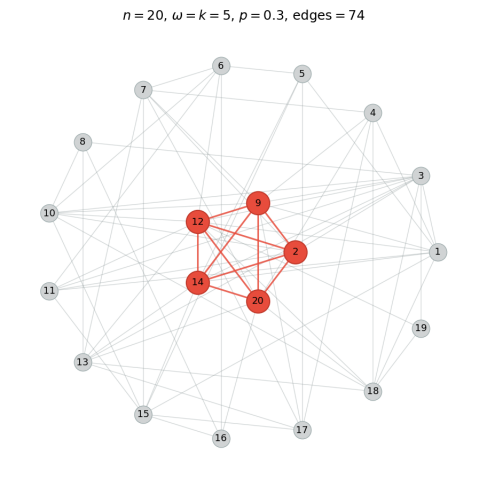
Six planted-clique StQP instances; for each, the clique is placed at three different matrix positions and each placement is submitted at four boson14 amplitudes.

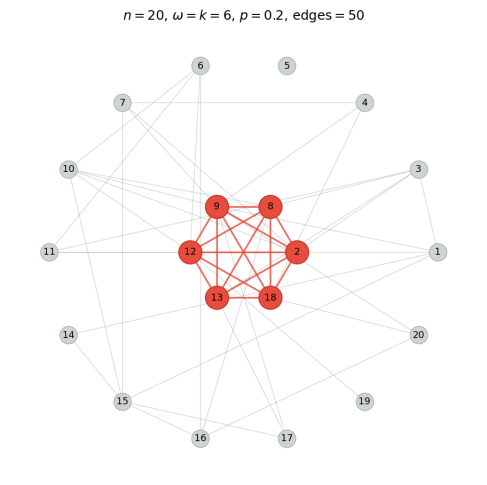
- **Variants** (index scrambling): *random* (original), *front* (clique at indices $1..k$), *end* (indices $n-k+1..n$).
- **Amplitudes**: $a \in \{300, 400, 500, 600\}$ (one instance uses $a = 380$ and another adds $a = 700$).
- **Samples per run**: 100. Sum constraint $R = 100$.

instance	n	k	p	g^*
n20_k5	20	5	0.3	4000.0
n20_k6	20	6	0.2	4166.7
n50_k10	50	10	0.3	4500.0
n50_k22	50	22	0.5	4772.7
n100_k22	100	22	0.2	4772.7
n100_k63	100	63	0.7	4920.6

Seed $= 42$, $R = 100$ for all instances.

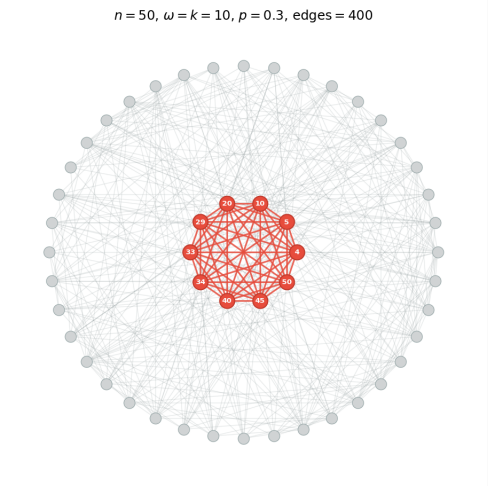
Test graphs: $n = 20$ (planted clique highlighted)

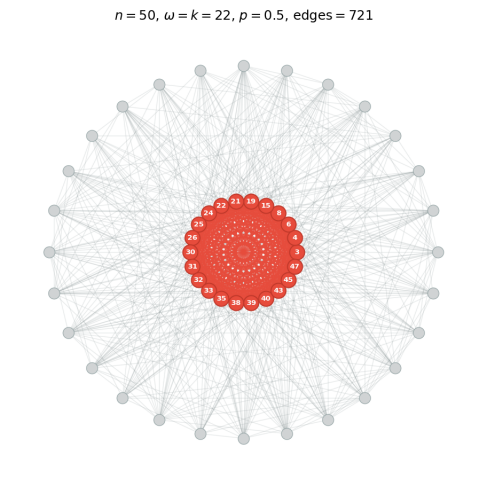




Red nodes = planted maximum clique (brute-force verified $\omega = k$).

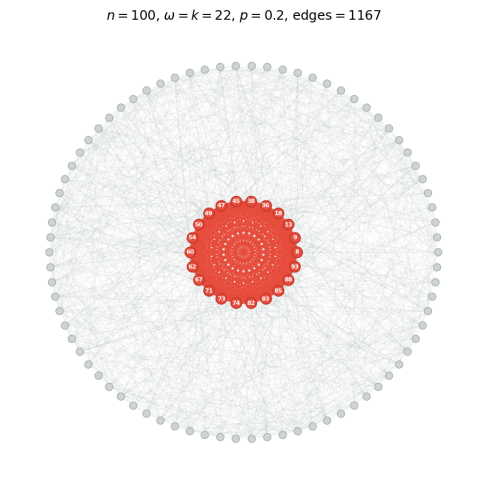
Test graphs: $n = 50$ (planted clique highlighted)

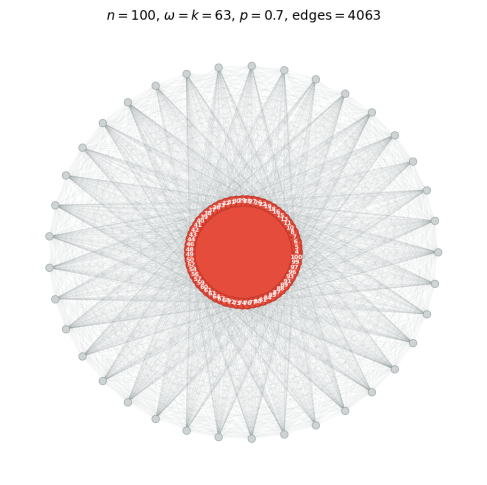




Inner ring = clique nodes, outer ring = remaining vertices.

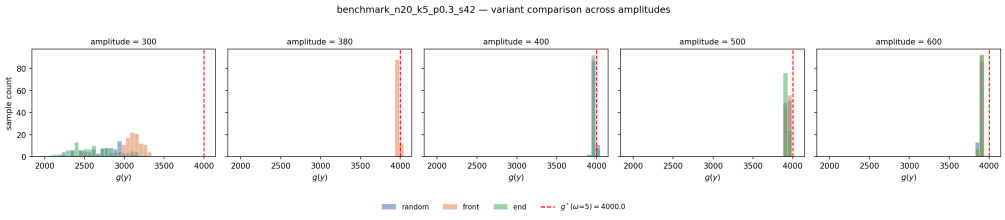
Test graphs: $n = 100$ (planted clique highlighted)





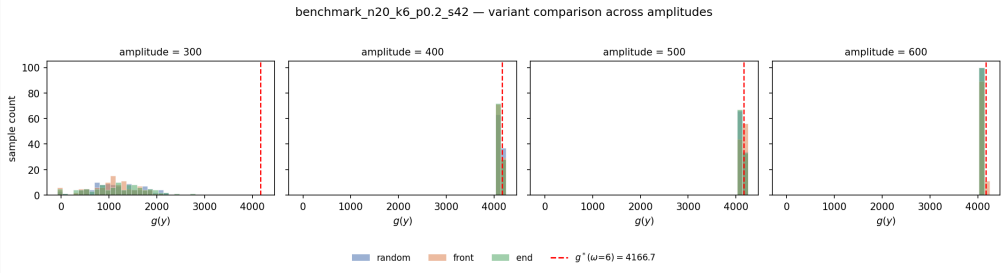
Clique-only labels shown to keep the figure legible at $n = 100$.

Results: n20_k5 ($k = 5$, $g^* = 4000$)



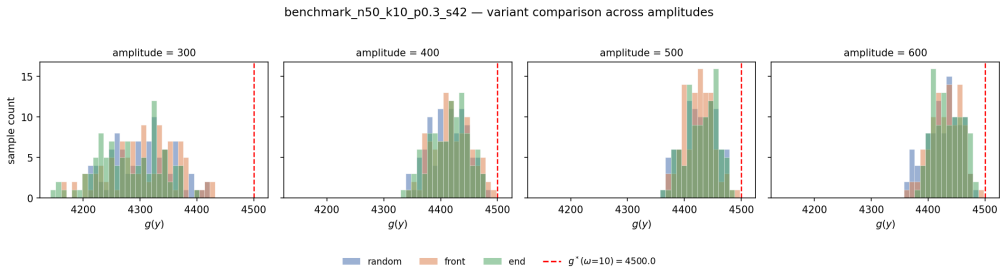
Best sample hits 99.93% of g^* (front, $a = 380$); omega-hit rate reaches 100% at $a \in \{400, 500\}$. Variants overlap almost perfectly.

Results: n20_k6 ($k = 6$, $g^* = 4167$)



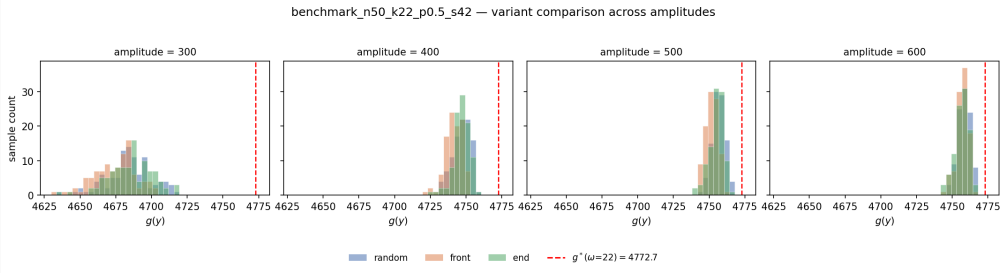
Best sample hits 99.95% of g^* (front, $a = 500$); omega-hit rate peaks at 100% for all variants at $a = 500$, drops at $a = 600$.

Results: n50_k10 ($k = 10$, $g^* = 4500$)



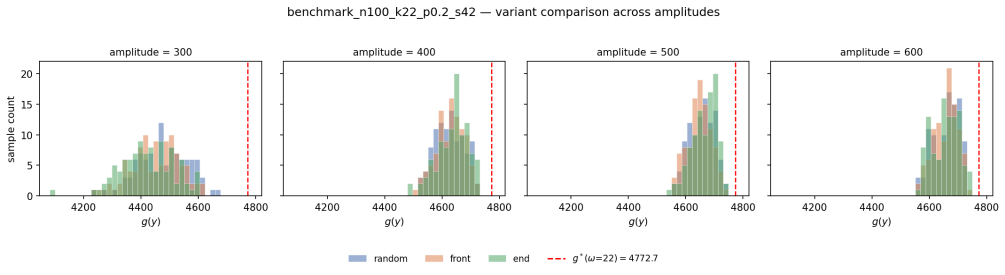
Best sample hits 99.86% of g^* (front, $a = 500$). Distributions now concentrate just below g^*; omega-hit rate collapses to $\leq 7\%$ despite the gap being under 0.3%.

Results: n50_k22 ($k = 22$, $g^* = 4773$)



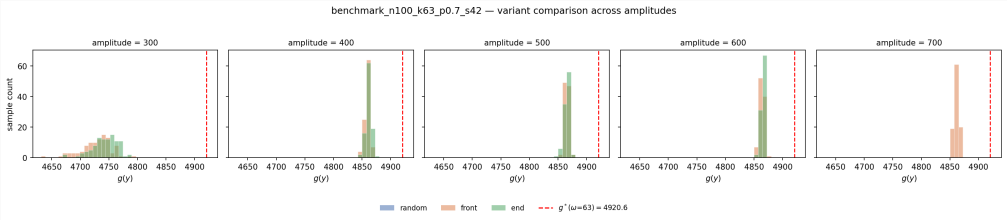
Best sample hits 99.88% of g^* (random, $a = 600$). Omega rounding never lands exactly at $k = 22$ — integer recovery fails at this resolution.

Results: n100_k22 ($k = 22$, $g^* = 4773$)



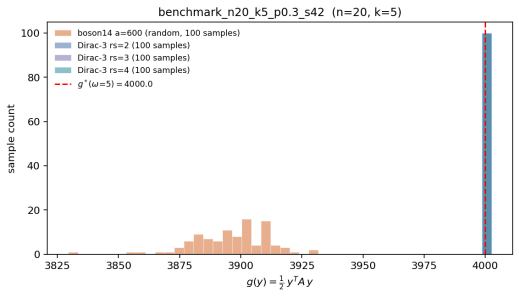
Best sample hits 99.41% of g^* (front, $a = 600$). Larger n shifts the entire distribution slightly further below the MS line.

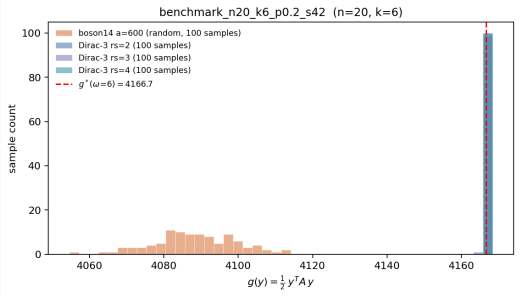
Results: n100_k63 ($k = 63$, $g^* = 4921$)



Densest graph ($p = 0.7$). Best sample hits 99.06% of g^* (end, $a = 400$). *Random* variant missing — only *front*/*end* were submitted for this instance; *front* has an extra $a = 700$ run.

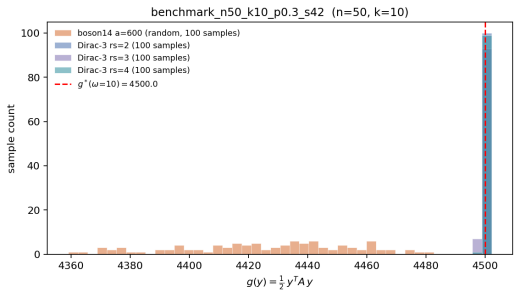
Dirac-3 vs boson14 (a=600): $n = 20$

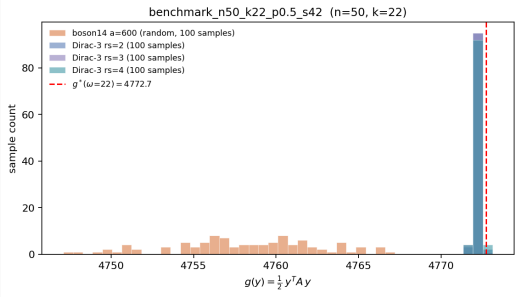




Each Dirac run: 100 samples at relaxation-schedule $\in \{2, 3, 4\}$. On small graphs Dirac-3 locks onto g^*; boson14 at its best amplitude still sits a few percent below.

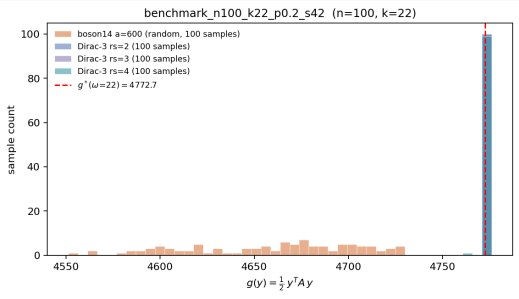
Dirac-3 vs boson14 (a=600): $n = 50$

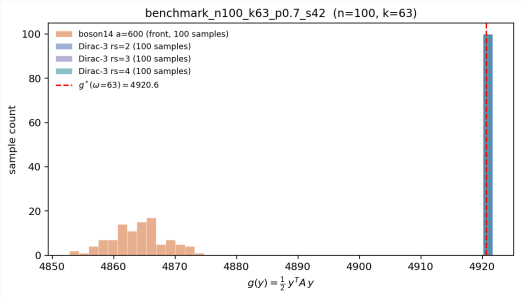




Dirac-3 and boson14 become more comparable as n grows. Schedule effect on Dirac is small; both solvers sample within a narrow band below g^*.

Dirac-3 vs boson14 (a=600): $n = 100$





Largest instances. Dirac-3 rs=4 gives the tightest high-g cluster; boson14 $a = 600$ histogram is comparable in spread but offset. MS line at g^*.

Sanity check: do Dirac-3 solutions correspond to real cliques?

For each of the 18 runs, take the best sample y, threshold the support at $y_i > R/(2k)$, and verify: **(i)** $|S| = \omega$, **(ii)** every pair in S is an edge of G, **(iii)** $y_i \approx R/\omega$ on S.

instance	rs	$\lvert S\rvert$	clique?	matches planted	y_S mean $\pm$ std (ideal R/ω)
n20_k5	2/3/4	5/5	✓	*alt clique*	18.8 ± 2.3 (20.0)
n20_k6	2/3/4	6/6	✓	yes	16.67 ± 0.09 (16.67)
n50_k10	2/3/4	10/10	✓	yes	10.00 ± 0.15 (10.00)
n50_k22	2/3/4	22/22	✓	yes	4.545 ± 0.12 (4.545)
n100_k22	2/3/4	22/22	✓	yes	4.545 ± 0.09 (4.545)
n100_k63	2/3/4	63/63	✓	yes	1.587 ± 0.04 (1.587)

All 18 supports are valid max cliques. The *n20_k5* graph happens to contain 4 distinct 5-cliques; Dirac consistently landed on $\{2, 3, 9, 12, 14\}$ instead of the planted $\{2, 9, 12, 14, 20\}$ — both have $|S| = \omega = 5$, so MS objective is indifferent. The wider std on its support reflects proximity to a convex combination of overlapping optima.

Repairing boson14 samples

Apply the same threshold $y_i > R/(2k)$ to every boson14 sample to extract a support S. Three derived quantities per sample:

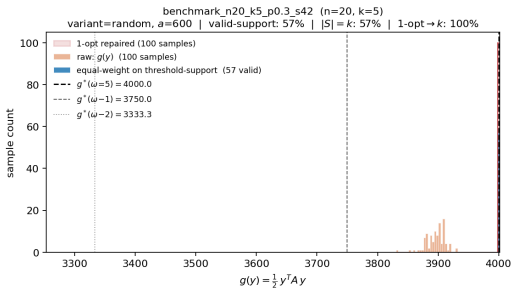
- **raw** — $g(y) = \frac{1}{2} y^\top A y$ on the continuous vector;
- **equal-weight** — when S is a clique, the MS-optimal embedding on it has $g = \frac{R^2}{2}(1 - 1/|S|)$;
- **1-opt repaired** — seed the support, iteratively remove the worst-connected vertex until S is a clique, then greedily add any vertex adjacent to all of S until maximal. Report equal-weight g.

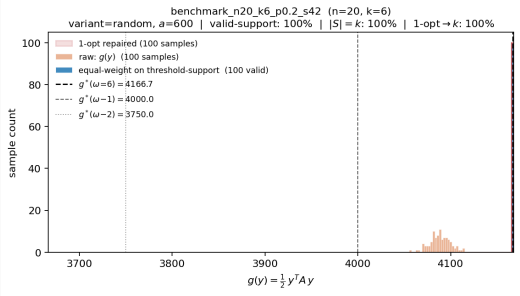
Aggregate over all 6 900 boson14 samples (6 graphs $\times$ 3 variants $\times$ up to 4 amplitudes $\times$ 100):

	runs (of 69)	samples (of 6 900)
Threshold-support is itself a clique	57 runs 100%	6 477 / 6 900 (93.9%)
1-opt repair reaches exactly ω	65 runs 100%	6 527 / 6 900 (94.6%)
1-opt reaches ω at $a = 600$ only	**all 17**	**1 700 / 1 700 (100%)**

At $a = 600$ (best amplitude), 1-opt recovers ω for **every single sample on every graph**. The failures are concentrated at $a = 300$ where the continuous output has too little mass on the right vertices for greedy extension to find them.

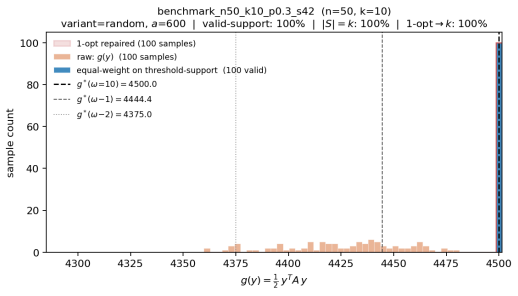
Repair overlays: $n = 20$

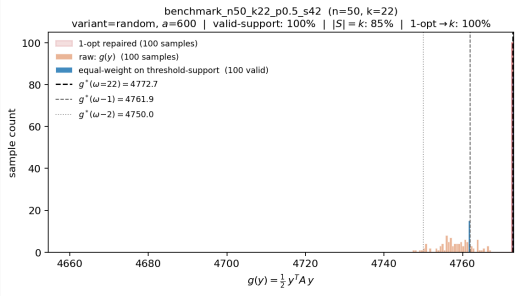




Orange: raw boson14 at $a = 600$. Blue: equal-weight MS value on samples whose threshold-support is already a clique. Faint red outline at g^*: all 100 samples after 1-opt repair.

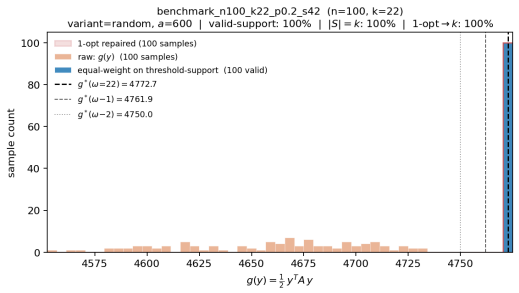
Repair overlays: $n = 50$

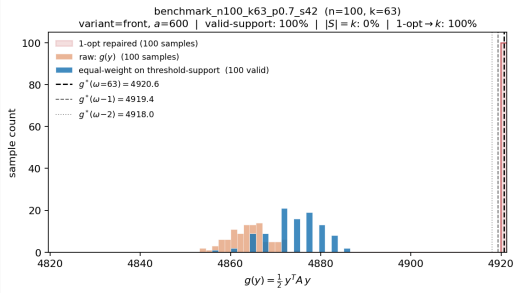




Raw boson14 sits a few percent below g^*; 1-opt moves *every* sample to the MS optimum. On `n50_k10` the threshold-support is already size-k and valid in all 100 samples.

Repair overlays: $n = 100$

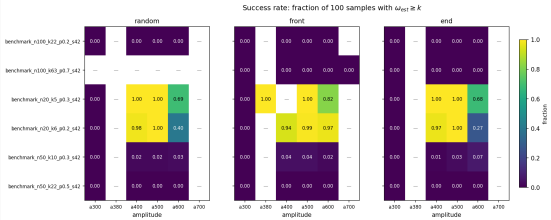




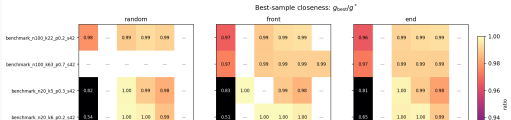
Largest graphs. The repair gap (raw $\rightarrow g^*$) is visually dramatic on `n100_k63`, but 1-opt still recovers the full 63-clique in every sample. The continuous output contains enough structural information even when the raw g is 1% below optimal.

Aggregate: success rate and best-sample closeness

fraction of 100 samples with $\omega_{\text{est}} \geq k$



g_{best}/g^* (scale 0.9–1.0)

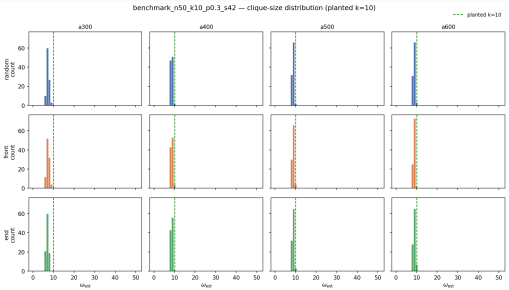


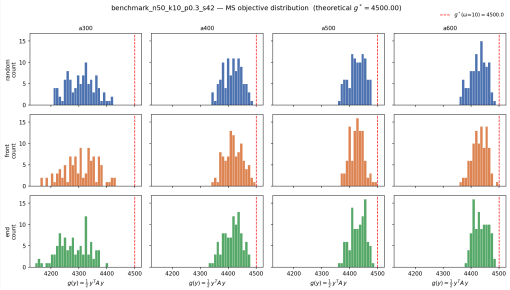
Left: exact omega recovery is reliable only for $k \leq 6$. Right: boson14 reaches $\geq 99\%$ of g^* on *every* amplitude except $a = 300$, where small-n runs fail to converge (0.51–0.83).

Findings

- **Clique-index permutation does not affect performance.** Across all instances the best g across variants differs by at most **0.24**% of g^* (largest gap: n50_k10). "Front", "end", and "random" placements are indistinguishable.
- **Amplitude is the dominant knob.** $a = 300$ systematically under-converges (best g/g^* as low as 0.51 on small instances); $a \in \{400, 500, 600\}$ all cluster at $\geq 99\%$.
- **Size n scales gracefully.** Best ratio drops only from ~ 1.00 at $n = 20$ to ~ 0.99 at $n = 100$.
- **Data note.** Instance `n100_k63` lacks the *random* variant; `n20_k5` used $a = 380$ instead of $a = 400$ for its *front* variant.

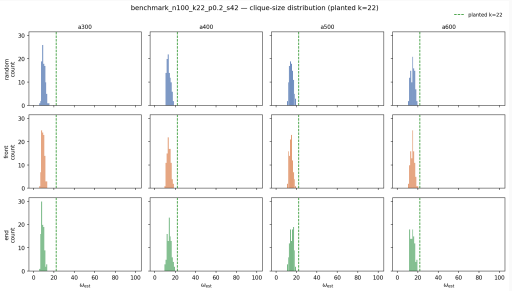
Appendix: `n50_k10` — clique-size and objective grids

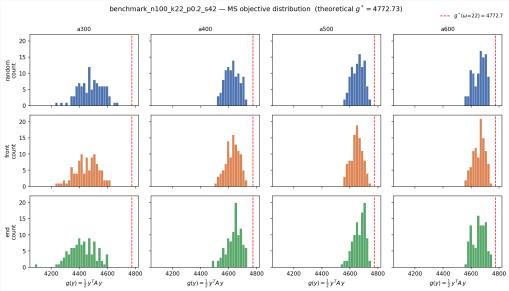




Left: ω_{est} mass concentrates at 9 even though best g is within 0.15% of $g^*(k{=}10)$. Right: objective distribution shifts towards g^* as amplitude rises.

Appendix: n100_k22 — clique-size and objective grids





Larger instance: ω_{est} spreads over 17–21; objective distributions still tight and consistent across variants.